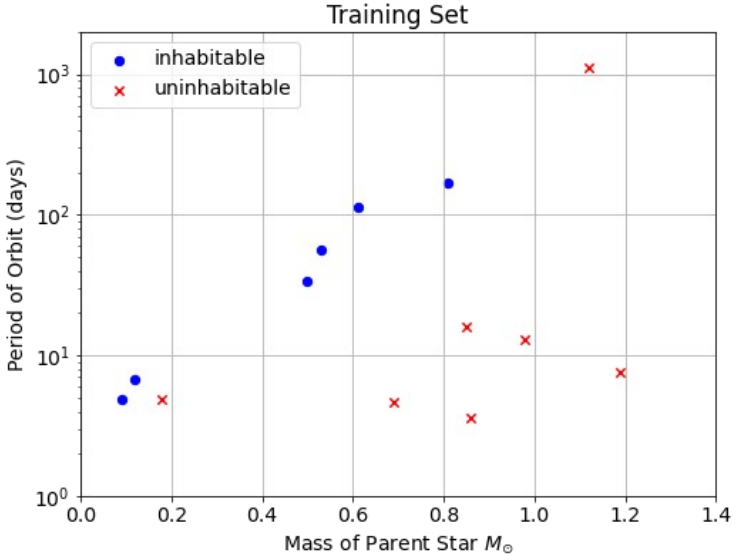


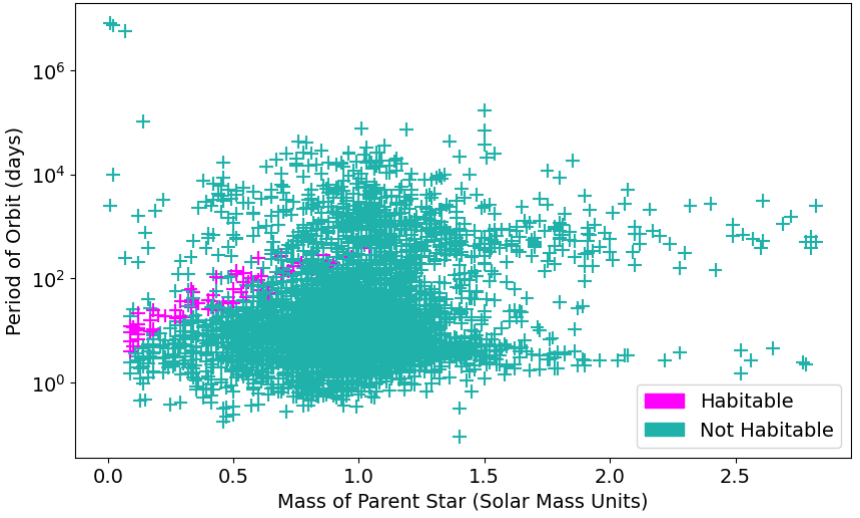
Diagnostics and Evaluation of Supervised Learning

Lecture/Lab 24
PHYS 4730 F26

Previously

- Two supervised learning algorithms
  - Decision Tree classifier
  - Nearest Neighbor classifier
- Idealized vs real datasets
  - Incomplete data
  - Features and labels need to be chosen and adjusted
  - Removal of outliers
  - Imbalance between labels



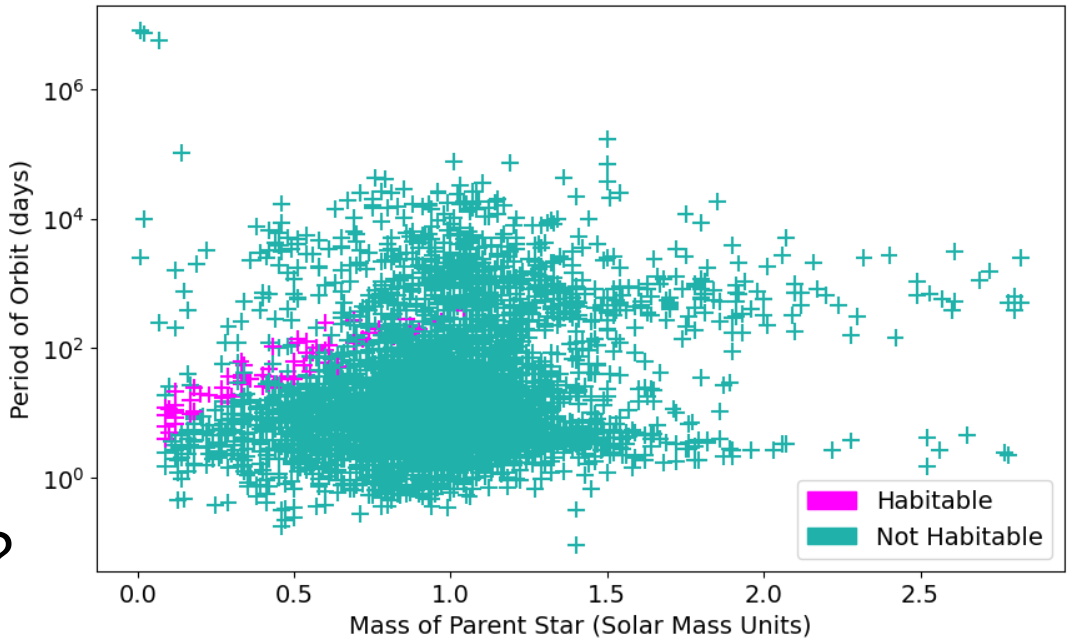


Today: Diagnostics

- Accuracy score
- TP/TN/FP/FN
- Precision Score
- Recall Score
- Confusion Matrix
- Learning curves

Accuracy Score

- Fraction of correct classifications
- For training set:
  - Ideally 100%
  - DT vs kNN
- Test set
  - Close to 100% = good
  - What about imbalanced data?



DT Test Accuracy Score: 0.99251
“Lazy” test accuracy score: 0.98801

Binary Classification: Four Outcomes

- Define:
  - One label := “positive”
  - One label := “negative”
- Relevant quantities in a classification are:
  - TP – True Positives
  - TN – True Negatives
  - FP – False Positives
  - FN – False Negatives
- “Confusion Matrix” CM

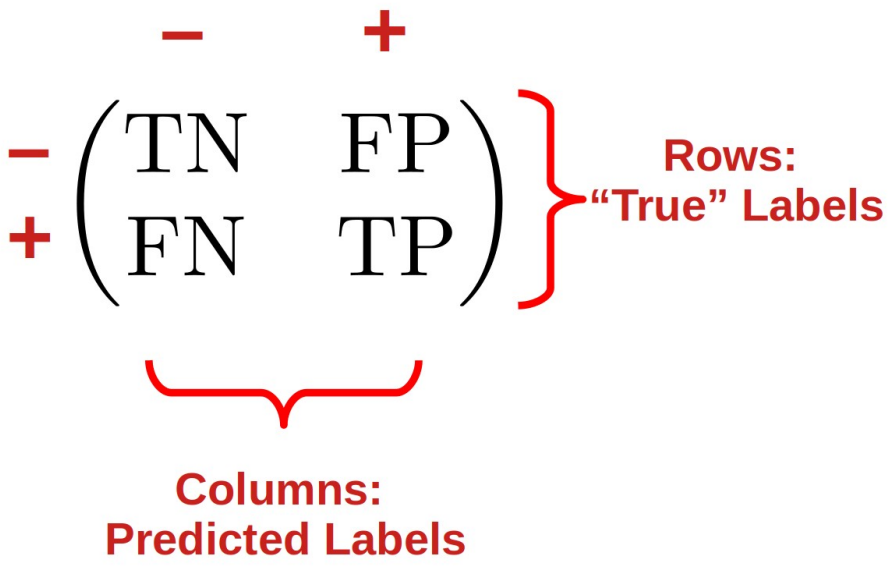
$$\mathrm{CM} = \begin{array}{c} \\ - \\ + \end{array} \begin{array}{c} \begin{array}{cc} - & + \end{array} \\ \begin{pmatrix} \mathrm{TN} & \mathrm{FP} \\ \mathrm{FN} & \mathrm{TP} \end{pmatrix} \end{array}$$

Rows: “True” Labels

Columns: Predicted Labels

Accuracy Score

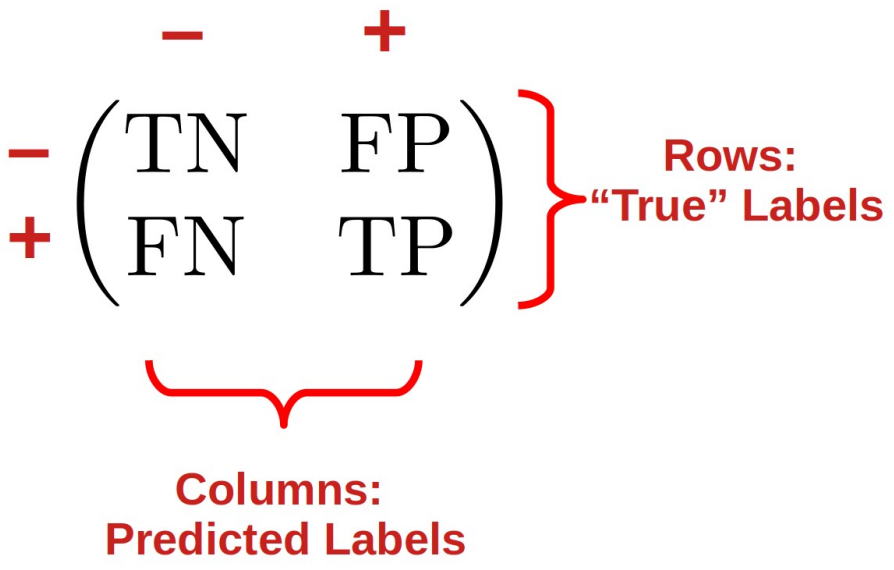
$$\text{accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{TP} + \text{FN} + \text{FP}}$$



- Symmetric in two error types
- Higher fractional misclassification for rarer class
- Less useful for unbalanced data

Precision Score

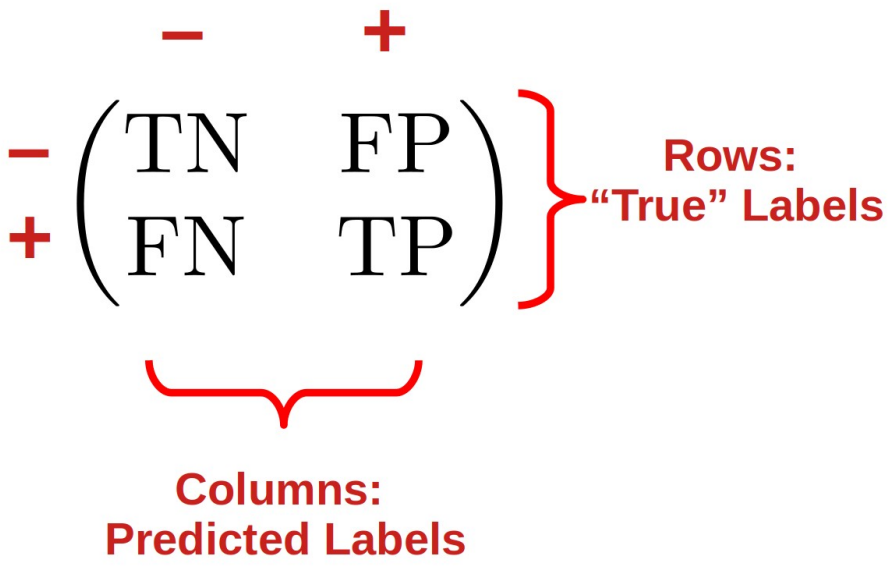
$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$



- Fraction of positive identifications which are correct.
- Useful when we want high “purity”
- 1 – precision = contamination

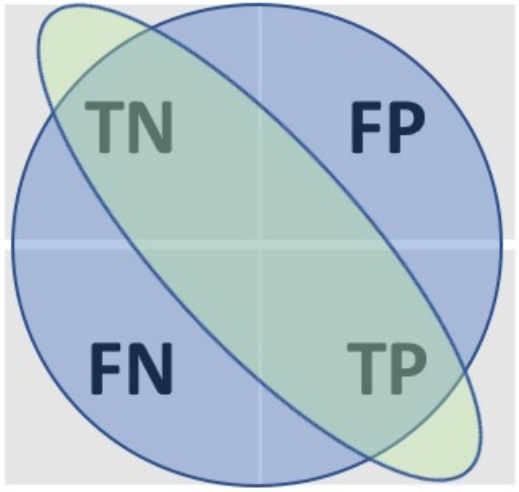
Recall Score

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

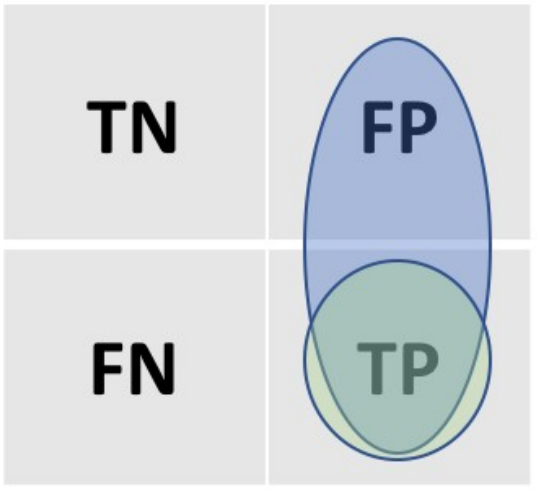


- Fraction of correctly identified positive examples.
- “completeness”
- If we don’t want to miss rare events.

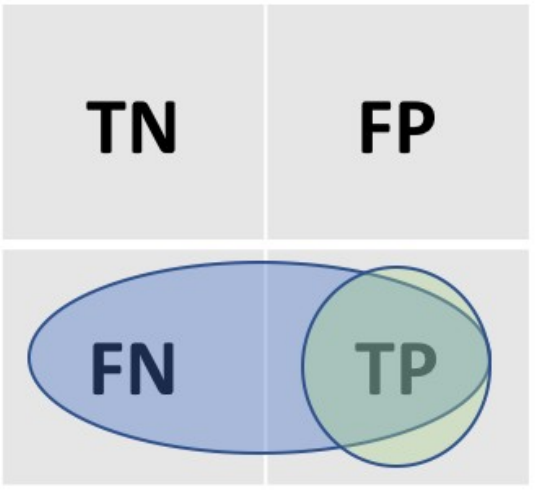
Mnemonic: Divide green/blue



accuracy



precision



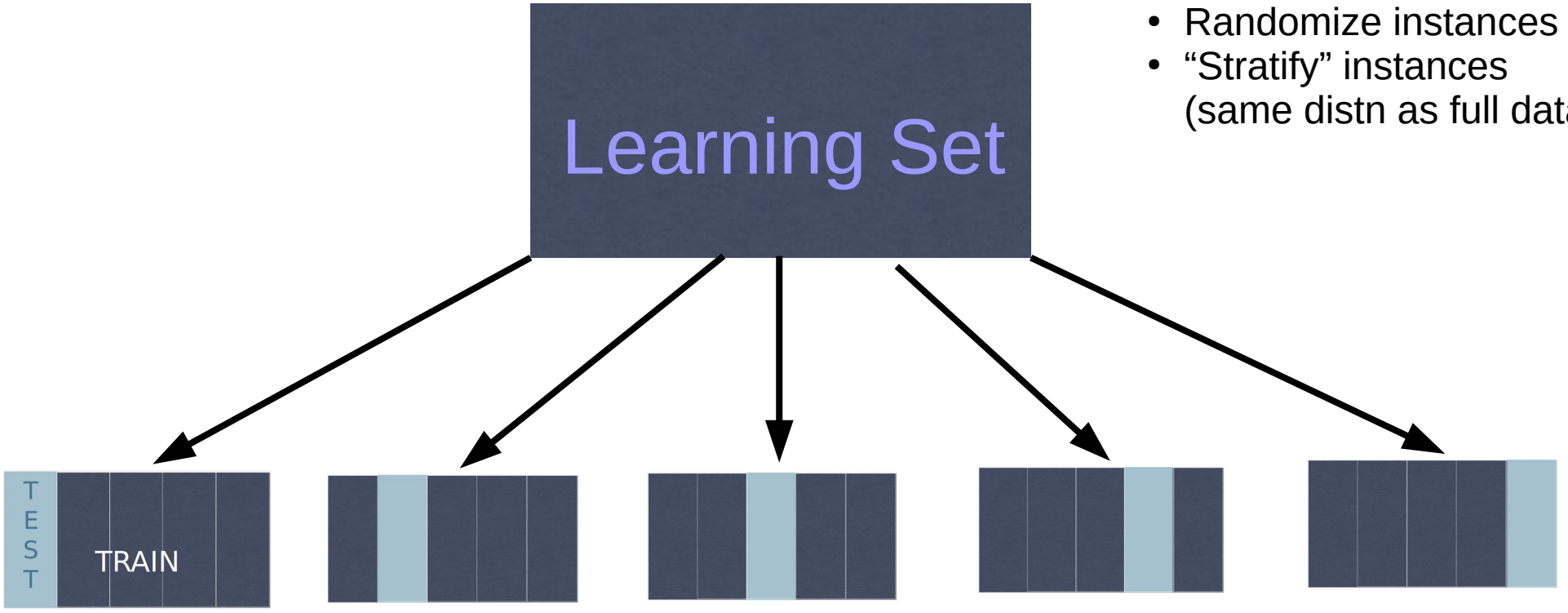
recall

Cross Validation

- Problem: Uncertainty due to statistical fluctuations in train/test split.
- Would be good to know:
  - “Typical” performance
  - Estimate of uncertainty
- Solution: Do the train/test split several times, note mean and standard deviation of the results.

Validation schemes:

- Divide data by 5
- Randomize instances
- “Stratify” instances (same distn as full data)



Splits for 5-fold cross validation

notebook: diagnostics.ipynb

Lab 24

Exercise 1.

In lecture, the instructor introduced the notebook diagnostics.ipynb as a survey of diagnostic tools in supervised learning. The notebook applied the technique of **k-fold cross validation** to the decision tree classifier of the full Habitable Worlds Catalog data set.

Starting with this notebook, experiment with the following:

- In addition to the 5-fold cross validation try varying *k* from 5 to 10. Note any differences.
- For the 5-fold case, try the alternative “shuffled” and “stratified” cross-validation schemes.

```
cv2 = KFold(shuffle = True, n_splits = 5, random_state=5)
cv3 = StratifiedKFold(shuffle = True, n_splits = 5, random_state=5)
```

  Again, note differences in the scores and learning curves.

Finally, in a markdown cell at the end of your notebook, please answer the following questions:

- If the accuracy score is 100%, what will be the precision and recall scores?
- What is the effect of shuffling and stratification on the k-fold splits?
- Within the particular k-fold schemes, are the differences in scores statistically significant for any of the splits? Explain.
- Does the confusion matrix depend on the cross-validation scheme chosen? Why or why not?
- Do the learning curves exhibit any significant differences for the three cross-validation schemes?